

Subject Description Form

Subject Code	EE4006 / EE4006A / EE4006B
Subject Title	Individual Project
Credit Value	6
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: The student should have completed most of the subjects required in previous years of the programme before taking this subject.
Objectives	To provide an opportunity for students: 1. to apply specialized professional engineering knowledge independently in the creative design, implementation, managing and evaluation of an engineering project, and 2. to identify key engineering problems, to solve them, and to communicate the findings in an oral and written report format.
Subject Intended Learning Outcomes	Upon completion of the subject, students will be able: a. To apply specialized knowledge independently. b. To identify key engineering problems, to solve them and to communicate what is achieved orally and in a written report. c. To develop a project which is creative, rich in intellectual content and sufficiently challenging. d. To monitor the progress of a project from concept to final implementation and testing, through problem definition and the selection of alternative solutions. e. To synthesize and apply their knowledge and analytical skills gained in various engineering domains. f. To build self-confidence, demonstrate independence, and develop professionalism by successfully completing the project in a competent manner.
Subject Synopsis/ Indicative Syllabus	<u>Phases of final-year project implementation</u> I. Project Selection Projects are proposed by staff or by an industrial partner. Projects may also be jointly proposed by student and staff. Industrial experience, research and consultancy activities are fertile ground for ideas. II. Initial Planning At the beginning of the project, students are required to conduct literature research to identify problems, define project objectives, and propose possible solutions. A preliminary schedule and budget must also be planned. All these elements need to be summarized in a proposal report. III. Project Implementation Throughout the project, students are required to maintain documentation of their progress and submit logbooks to their supervisor for review. IV. Mid-Project Review Midway through the project, students are required to submit an interim progress report and carry out a presentation to summarize their progress. This provides the supervisor and an assessor with a formal opportunity to assess the student's progress and address any issues if necessary.

V. Final Submission and Presentation

In Week 10 of the second semester, students must submit a comprehensive final report summarizing their work. This report will be assessed by the supervisor. To ensure it meets the required standards, students must first submit a draft to the supervisor for feedback before the final submission.

Students are also required to demonstrate and present their deliverables to an assessor and the project supervisor. This includes in-person presentations, production of demonstration videos, and a question-and-answer session.

Method of Assessment

In assessing the project, the supervisor and assessors will typically consider the following aspects:

- Intellectual achievement
- In-depth understanding of the topic and related subjects
- Quantity and quality of work done, including design and construction of equipment, experimentation, mathematical models, program writing, and verification
- Presentation skills, including the quality of the written report, the presentation itself, and responses to questions

All aspects of the project are thoroughly considered before determining the final grade. Examiners should take into account their experiences with the achievements of other projects in the Department from the current and previous years. The final-year project is assessed through 100% continuous assessment, composed of the following components:

I. Formal Project Proposal (5%)

The proposal should include:

- a. An abstract
- b. Introduction, including problem statement and objectives of the project
- c. Summary of the literature search
- d. Proposed specifications of the product
- e. Proposed approach/methodology to be used
- f. Some brief descriptions on the theory of the approach/methodology
- g. Preliminary time schedule of the project
- h. Cost estimation
- i. References

The proposal is assessed by the supervisor based on the following criteria:

1. *Overall Presentation*
2. *Abstract and Introduction*
3. *Literature Research*
4. *Methodology and Project Plan*

II. Interim Progress Report (10%)

The interim progress report should include:

- a. An abstract
- b. Introduction, including problem statement and objectives of the project
- c. Summary of the literature research
- d. Initial problem identification
- e. Preliminary suggestion on methodology
- f. Preliminary results and discussions
- g. Preliminary time schedule and milestones of the project
- h. Cost estimation
- i. References

This component is graded by the supervisor based on:

1. *Overall presentation*
2. *Abstract and introduction*
3. *Literature research*
4. *Methodology and technical skills*

5. *Results, discussions, and conclusions*

III. Interim presentation (10%)

Students are required to present their progress to an assessor after submitting the Interim Progress Report. The presentation will be assessed based on:

1. *Technical concepts*
2. *Demonstration and engineering accomplishment*
3. *Presentation skills*

IV. Final Report (30%)

The final project report should contain all works carried out by the student in the project. The length of the main body of the final report should be **at least 45 pages** in standard report format. Students are advised to form a framework for the report first, and then proceed to the formation of the titles of the chapters. The titles and structure of the sections within each chapter are then decided. Continuing the process, each section may be further expanded into appropriate sub-sections, divisions and sub-divisions etc., until a complete framework is formed. The content of the final report includes:

- a. An abstract
- b. Introduction, including problem statement and objectives of the project (especially any change from the original aims).
- c. Summary of the literature research
- d. The motivation behind the project and a brief outline of the project work.
- e. A summary of work done or developed in the project.
- f. The system design and the block diagram of the system, plus some brief descriptions on the theory.
- g. Results and discussion
- h. Difficulties encountered and the measures taken to solve them.
- i. The achievement of the project, the conclusions from the work, and suggestions for further work.
- j. A list of the references referred to the source of information in the report. This is compulsory.
- k. Materials which are closely related to the contents of the report, and which are themselves self-contained, may be included in the report as appendixes.

This component is graded by the supervisor based on:

1. *Overall presentation*
2. *Abstract and introduction*
3. *Literature research*
4. *Methodology and technical skills*
5. *Results, discussions, and conclusions*

V. Final Presentation and Demonstration (30%)

Students need to give **final presentations** to the supervisor and one assessor, demonstrating their mastery of the project topics by elaborating on technical details and providing satisfactory answers to questions. The presentation should be concise and engaging, utilizing effective visual aids and multimedia, maintaining a logical flow of ideas, and controlling the pace appropriately. Good pronunciation and intonation are desirable, and courtesy is expected throughout the presentation.

Demonstration will take place both during the presentation and through a produced video to showcase the hardware or software. Hardware must be neatly built, circuits and software should function correctly, and experiments should support the fulfillment of project objectives.

This component will be assessed by the supervisor and one assessor based on:

1. *Technical concept*
2. *Demonstration and engineering accomplishment*
3. *Presentation skills*

	VI. Continuous Assessment (15%) The 15% continuous assessment consists of a 5% logbook and 10% overall performance. The logbook serves as a documentation of regular project progress and must be endorsed by the supervisor. The overall performance will be assessed by the supervisor based on the following criteria: 1. <i>Motivation and perseverance</i> 2. <i>Originality and innovation of the project</i> 3. <i>Execution and problem-solving skills</i> 4. <i>Communication</i> 5. <i>Self-discipline and time management</i> 6. <i>Milestone reports</i> Note 1: Each student has to submit/carry out all the components (I to VI) before he/she is considered to have completed the FYP. Note 2: The final grade for the FYP will be calculated by taking the weighted average of the grades from the above six components.																																																																						
Teaching/Learning Methodology	As the nature of the subject implies, there will not be formal lecture in the subject, other than a few hours of briefings on general information, some procedures in project administration and some techniques on information/components searching. Students learn the technical contents by a substantial number of individual discussions with their project supervisors and a large number of hours of self-learning. The planning of the project will be conducted under the direction of the supervisor. Through the execution of the project plan with guidance from the supervisor, the student should be able to achieve the learning outcomes. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="6">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Discussion with the project Supervisor</td><td>✓</td><td></td><td>✓</td><td></td><td></td><td></td></tr><tr><td>Writing of the project proposal</td><td>✓</td><td>✓</td><td>✓</td><td></td><td>✓</td><td></td></tr><tr><td>Writing of the interim report</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Writing of the final report</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Presentation and demonstration</td><td></td><td>✓</td><td></td><td></td><td></td><td>✓</td></tr></table>	Teaching/Learning Methodology	Outcomes						a	b	c	d	e	f	Discussion with the project Supervisor	✓		✓				Writing of the project proposal	✓	✓	✓		✓		Writing of the interim report	✓	✓	✓	✓	✓		Writing of the final report	✓	✓	✓	✓	✓	✓	Presentation and demonstration		✓				✓																						
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Formal project proposal</td><td>5%</td><td></td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>2. Interim progress report</td><td>10%</td><td></td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>3. Interim presentation</td><td>10%</td><td></td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td></tr><tr><td>4. Final report</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>5. Presentation and demonstration</td><td>30%</td><td>✓</td><td>✓</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>6. Continuous assessment</td><td>15%</td><td>✓</td><td></td><td></td><td>✓</td><td></td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Assessment criteria for each of the above assessment methods are as listed in one of above sections.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						a	b	c	d	e	f	1. Formal project proposal	5%		✓	✓				2. Interim progress report	10%		✓	✓	✓			3. Interim presentation	10%		✓		✓		✓	4. Final report	30%	✓	✓	✓	✓	✓	✓	5. Presentation and demonstration	30%	✓	✓				✓	6. Continuous assessment	15%	✓			✓		✓	Total	100%						
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Student Study Effort Expected	Class contact:	
	▪ Briefings	3 Hrs.
	▪ Individual discussions with supervisor	36 Hrs.
	Other student study effort:	
	▪ Information search, self-study, execution of the project, report writing, preparation of presentation	171 Hrs.
	Total student study effort	210 Hrs.
Reading List and References	To be advised by supervisor	

July 2025